

Clonal analysis for understanding fate biases in developing embryos

Sergey Isaev¹ and Igor Adameyko^{1,2}



Multipotent progenitors that appear phenotypically similar often differ in the range of cell types they generate, which is typically described as fate bias. Traditional bulk lineage-tracing approaches, such as Cre/Lox systems, established qualitative progenitor–progeny relationships but offered limited insight into mechanisms governing tissue composition. Recent advances in lineage tracing combined with single-cell transcriptomics enable comprehensive characterization of clonal diversity at the whole-embryo scale, promising to provide mechanistic insights into developmental robustness. This review summarizes the current state of the art experimental and computational approaches in the field, with emphasis on emerging spatial and perturbation-based strategies in clonal biology.

Addresses

¹ Department of Neuroimmunology, Center for Brain Research, Medical University Vienna, Vienna 1090, Austria

² Department of Physiology and Pharmacology, Karolinska Institutet, Stockholm 17177, Sweden

Corresponding authors: Isaev, Sergey (sergey.isaev@meduniwien.ac.at), Adameyko, Igor (igor.adameyko@meduniwien.ac.at), Adameyko, Igor (X <https://x.com/adameykolab>)

Current Opinion in Genetics & Development 2026, **99**:102475

This review comes from a themed issue on **Developmental Patterning and Evolution**

Edited by **Seth Blackshaw** and **Christian Mayer**

For complete overview of the section, please refer to the article collection, “**Developmental Patterning and Evolution (2026)**”

Available online 21 April 2026

<https://doi.org/10.1016/j.gde.2026.102475>

0959–437X/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

During development, cells undergo progressive differentiation, a process classically illustrated by Waddington’s epigenetic landscape. This metaphorical topography maps the state space of potential cellular identities, where ‘valleys’ (stable attractors) and ‘ridges’ (unstable barriers) govern state transitions [1] (Figure 1a). Cells move across this landscape driven by sequential shifts in gene expression, where complex positive and negative feedback loops between different gene regulatory networks (GRNs) sculpt specific differentiation trajectories [2]. One of the central questions in developmental biology

remains the molecular logic underlying cell fate choice — the bifurcations in the landscape where a cell must ‘decide’ between divergent paths. Identification and interpretation of this logic is fundamental to our understanding of development and is a prerequisite for the rational engineering of tissues and organoids [3].

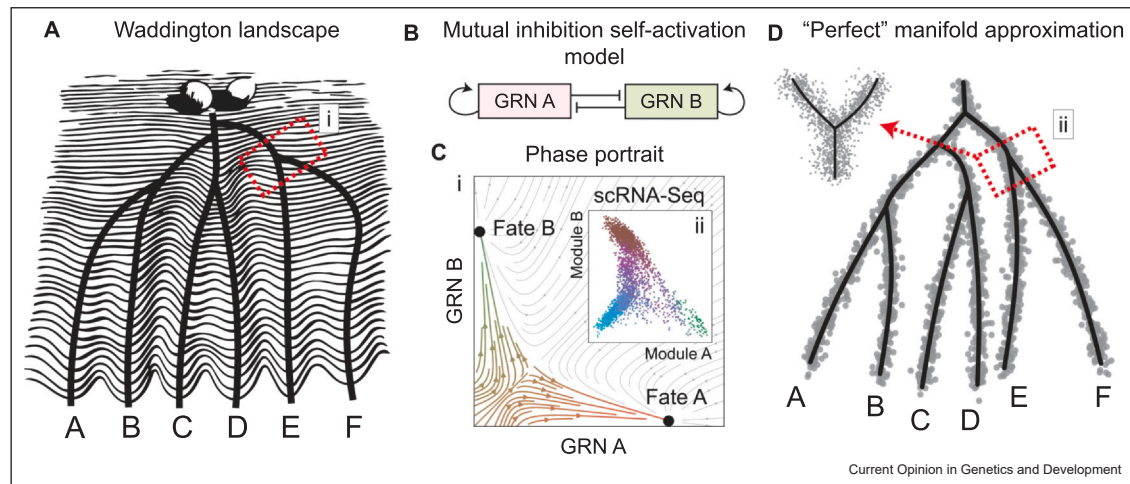
Several mechanisms have been proposed to explain lineage selection [5,6]. One prominent model describes fate decisions as a bifurcation resulting from competition between two antagonistic GRNs [7] (Figure 1b). This can be distilled into a three-step mechanism [8]:

1. **Priming:** Multipotent progenitors enter a state where competing GRNs for alternative fates are co-expressed at low or fluctuating levels.
2. **Biasing:** External signals or stochastic fluctuations tilt this equilibrium, rendering one GRN slightly dominant and shifting downstream fate probabilities without full commitment.
3. **Lock-in (Commitment):** Once a critical threshold is crossed, positive feedback and mutual inhibition solidify the choice, irreversibly committing the cell to a specific lineage.

This mechanism is a direct application of formal dynamical systems theory in developmental biology [9] (Figure 1c(i)) and has been experimentally validated across diverse biological contexts [7,10].

The advent of single-cell RNA sequencing (scRNA-Seq) has enabled the simultaneous measurement of tens of thousands of individual transcriptomes, allowing us to reconstruct molecular portraits of fate decisions at unprecedented resolution. The distribution of cells in high-dimensional gene expression space provides a robust approximation of the Waddington landscape [11] (Figure 1d). Notably, the transcriptional dynamics observed prior to bifurcation align closely with the mutual inhibition/self-activation model predictions [8] (Figure 1c(ii), from [4]). A critical implication of this model is that lineage biasing can occur long before overt differentiation markers appear; seemingly homogeneous progenitor cells may harbor distinct biases based on the subtle stoichiometric balances of competing GRNs [12,13]. This observation motivated the development of computational frameworks like FateID [14] (Figure 2a), which predicts fate biases by analyzing the activity of competing gene signatures in progenitor populations.

Figure 1



From Waddington's landscape to scRNA-Seq manifold. **(a)** The Waddington landscape (from [1] with modifications) showing average trajectories of differentiation into six cell types. The bifurcation (i) is highlighted in red. **(b)** Mutual inhibition/self-activation model with two competing GRNs. **(c)** (i) Phase portrait corresponding to the model in (b), and (ii) a scatter plot (from [4]) showing the activity of two competing gene modules per cell in scRNA-Seq data undergoing a bifurcation. **(d)** 'Perfect' manifold approximation of the Waddington landscape from scRNA-Seq data. The bifurcation (ii) is highlighted in red.

While signature-based approaches are intuitive, the non-linear dynamics of GRNs often limit their predictive power. Consequently, methods such as MIOFlow [15] and Waddington-OT [16] utilize the geometry of the scRNA-Seq manifold itself to estimate fate biases, often employing optimal transport (OT) to model transition dynamics (Figure 2b). However, to resolve the directionality of movement, these methods typically require external temporal 'guidance', such as experimental timepoints or prior knowledge of progenitor–progeny relationships. The resulting models are thus highly sensitive to the underlying assumptions regarding manifold transitions [17].

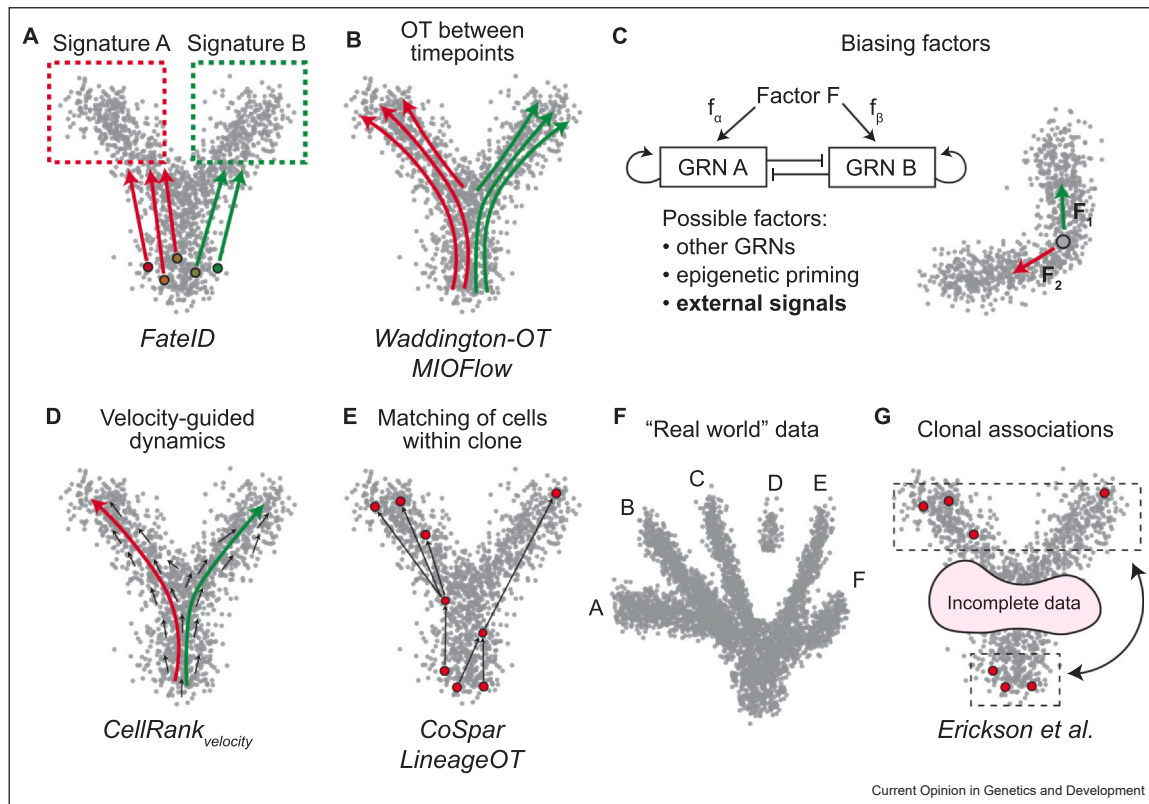
Beyond these methodological sensitivities, differentiation is also orchestrated by extrinsic factors — including signaling contexts and spatial niche constraints [18] — and epigenetic priming [19] (Figure 2c). This can lead to 'hidden variable' scenarios where cells with identical transcriptomic profiles exhibit divergent differentiation dynamics [17], a phenomenon invisible to purely transcriptomic manifold methods. RNA velocity [20] emerged as a promising tool to infer directionality by analyzing the ratio of unspliced to spliced mRNA transcripts. In theory, this vector reflects the net effect of internal and external forces to predict the future state of individual cells. Methods like CellRank incorporate RNA velocity to model movement across the manifold, significantly enhancing fate prediction accuracy [21] (Figure 2d). Nevertheless, RNA velocity is subject to technical limitations and biological assumptions that can limit its reliability in complex systems [22,23].

Importantly, CellRank and related frameworks project velocity vectors onto a k-nearest-neighbor graph to compute cell-to-cell transition probabilities, meaning that the manifold topology directly shapes the resulting fate predictions even when velocity itself is accurate.

The limitations of purely transcriptomic manifolds have motivated the coupling of scRNA-Seq with lineage tracing [24]. While tracing does not capture continuous state transitions, it provides the essential 'ground truth' of cellular history in the form of the final fate distribution of a given cell's progeny. Lineage recording systems generally fall into two categories: evolving barcodes (e.g. CARLIN [25], DARLIN [26], scGESTALT [27], LINNAEUS [28], ScarTrace [29]), which utilize Clustered Regularly Interspaced Short Palindromic Repeat-induced scars to reconstruct deep phylogenies, and static barcodes (e.g. CellTagging [30], LARRY [13], TREX [31]), which label progenitor pools with unique, heritable sequences. For the purpose of fate-mapping, static barcodes are often preferred, as evolving clones can be computationally collapsed into static 'clades' for analysis.

In vitro systems further enable 'sister cell' assays, where a clone is split: one 'arm' reveals the progenitor's transcriptomic state while the other reveals its eventual fate. Computational frameworks like CoSpar (graph-regularized optimization) [32] and LineageOT [33] bridge these modalities to clarify progenitor dynamics based on progeny distributions (Figure 2e). Interestingly, despite the inclusion of clonal data, these methods often exhibit predictive accuracies similar to velocity-guided models

Figure 2



Approaches for identifying biasing correlates in progenitors. **(a)** Signature-based methods (FateID) score expression signatures associated with alternative fates in progenitor cells. **(b)** Manifold-based OT methods (Waddington-OT, MIOFlow) find optimal matchings between cells at earlier and later timepoints. **(c)** Hidden factors — including other GRNs, epigenetic priming, and external signals — can produce divergent biasing outcomes from identical transcriptomic profiles (right), limiting purely manifold-based predictions. **(d)** Velocity-guided methods (CellRank) incorporate RNA velocity vectors to improve fate bias prediction. **(e)** Lineage-informed methods (CoSpar, LineageOT) use clonal relationships to link progenitor states to fate outcomes. **(f)** In practice, the manifold assumption is often violated in real datasets. **(g)** Manifold-free approaches directly test associations between progenitor gene expression and clonal fate distributions (Erickson et al.).

[34], likely reflecting inherent bottlenecks in the transcriptomic manifold approximation itself.

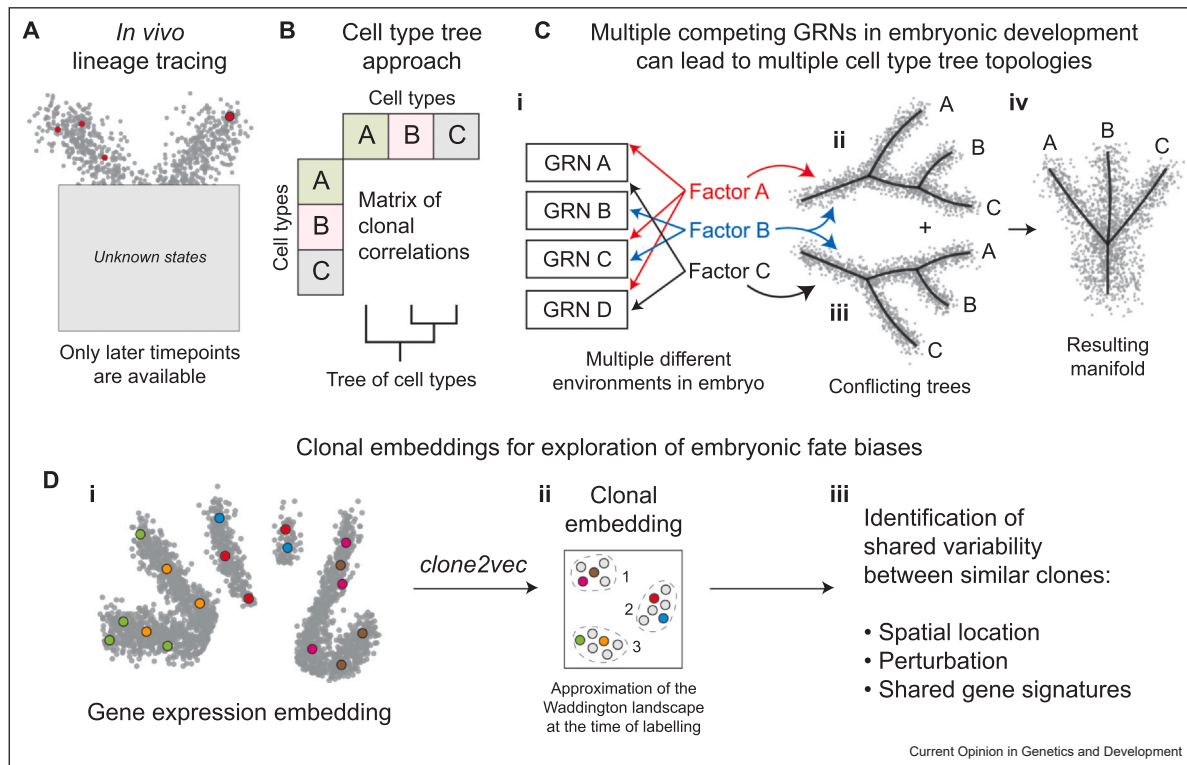
More broadly, however, the manifold is often an incomplete representation regardless of the analytical strategy. Measurement noise blurs the topography, leaving barrier heights and attractor depths poorly defined. Datasets are frequently sparsely sampled, missing critical intermediate states and violating the assumption of trajectory continuity (Figure 2f). The manifold assumption can be bypassed by directly testing associations between gene expression in the progenitor arm of a clone and the fate outcome of its differentiated arm (Figure 2g), but this comes with its own disadvantages — primarily the small number of clonally labeled cells in most datasets [13,32].

In the developing embryo, multi-timepoint clonal analysis remains technically elusive. With few exceptions where progenitors and progeny coexist — such as hematopoiesis

[35] or mesenchymal differentiation [36] — direct application of described approaches is often impossible (Figure 3a). As a result, many *in vivo* studies are restricted to reconstructing cell type differentiation trees based on the covariation of clonal abundances across cell types [37,38] and similar approaches [39,40] (Figure 3b) or must rely on computational inference from the non-barcoded data [41,42]. Furthermore, *in vivo* barcoding is significantly more complex than *in vitro* applications — often barcode injection and manipulation are performed *ex vivo* before re-implantation, which limits its utility for studying undisturbed embryonic development.

The described ‘differentiation tree’ approach assumes a fixed, universal structure [43], an assumption that may not hold *in vivo*. Differentiation is heavily regulated by spatial signaling contexts, meaning the topology of the differentiation tree can vary significantly across different embryonic locations (Figure 3c). To address this, the computational method clone2vec groups clones by their

Figure 3



In vivo lineage tracing and tree-free approaches to fate bias identification. **(a)** *In vivo* lineage tracing typically captures only later timepoints, with progenitor states unobserved. **(b)** The cell type tree approach infers differentiation hierarchies from a matrix of clonal co-occurrence across cell types. **(c)** In embryonic development, multiple competing GRNs are modulated by distinct spatially varying factors (i), which can produce conflicting cell type tree topologies in different embryonic regions (ii, iii). The resulting scRNA-Seq manifold (iv) averages over these contexts, obscuring location-specific branching structures. **(d)** The clone2vec strategy bypasses tree reconstruction entirely. Clones observed at a later timepoint (i) are embedded by their fate distributions (ii). Clusters of clones with similar fate profiles are then analyzed for shared features that may underlie the observed biasing patterns (iii).

fate distributions to identify shared molecular features across clones with similar behaviors [36]. This 'tree-free' method does not require the presence of progenitor cells in the dataset, relying instead on the assumption that shared features across distinct clones reveal underlying regulatory mechanisms (Figure 3d).

clone2vec is not a direct competitor for methods that aim to identify biasing programs in the progenitor population; indeed, some of those methods may have higher sensitivity for that specific task. Rather, clone2vec operates at a different level of abstraction. Clonal embedding — the output of the algorithm — represents each clone as a point in a space of fate probabilities under a multinomial assumption. It treats each clone as a random variable sampled from the 'true' clonal distribution in gene expression space, shifting the question from "what biases a particular clone toward one or another fate?" to the more general "which biasing patterns do we observe across the dataset?" — all without relying on discrete cell type assignments. The resulting clonal clusters can reveal stereotypical fate

distributions encompassing whole ensembles of cell types, which better reflects the reality of embryonic development, where borders between cell types are often fuzzy, while at the same time fine spatial regulation produces consistent proportions of cell types across different body regions. Variation of clonal patterns can uncover smooth or abrupt transitions in lineage composition across space, thereby revealing how local anatomical structures and boundaries emerge and providing insight into the fundamental principles of morphogenesis.

In line with this logic, a critical shared feature between clones with similar behavior — and a primary readout of signaling context — is spatial location [36]. Because most clones (excluding highly migratory populations like neural crest or immune cells) exhibit restricted movement [44], sister cells share a history of molecular and physical cues. Linking fate distribution to spatial coordinates represents a conceptual shift: rather than asking if a clone is biased, we ask if the tissue contains 'bias fields' — spatially patterned regimes that amplify,

suppress, or redirect intrinsic lineage tendencies. Recent advances in imaging-compatible RNA barcoding are now bridging the gap between high-resolution spatial maps and clonal identity [31,45].

Ultimately, both lineage tracing and RNA velocity provide correlates of fate biases rather than causes. Moving from descriptive to causal models requires the interpretation of GRN activity and its modulation by molecular signals and ‘master’ transcription factors. Several computational methods have been developed to predict cellular responses to perturbations as a proxy for fate transitions. Benchmarking studies suggest that CellOracle [46], a regression-based approach, currently outperforms other methods in this task [47]. However, the most definitive way to identify the drivers of fate bias is through systematic perturbations. Although analyzing tissue composition following a perturbation provides a direct readout of its influence on fate decisions, an additional clonal layer of information can give insights into the level of heterogeneity in mechanisms behind the bulk compositional effect, opening doors to further explanation of the observed effects [36].

The combination of spatial transcriptomics, lineage tracing, and mosaic perturbation screening (particularly for ligand-receptor interactions) represents the next frontier in developmental biology. Together, these modalities can map the bias fields introduced above — the exact biasing patterns triggered by specific signals within their native spatial contexts — closing the loop between correlative fate prediction and mechanistic understanding.

CRedit authorship contribution statement

Sergey Isaev: Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Igor Adameyko:** Conceptualization, Visualization, Writing – original draft, Writing – review & editing, Project administration, Supervision, Funding acquisition.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

SI and IA were supported by the ERC Synergy grant (KILL-OR-DIFFERENTIATE), Swedish Research Council, Paradiance Foundation, Cancer Foundation in Sweden, Knut and Alice Wallenberg Foundation, Austrian Science Fund (consortium SFB F78 program and Emerging Fields “Brain Resilience” consortium, as well as Stand-Alone Project Grants).

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
- of outstanding interest

1. Waddington CH: **The Strategy of the Genes**. Routledge; 2014.
2. Moris N, Pina C, Arias AM: **Transition states and cell fate decisions in epigenetic landscapes**. *Nat Rev Genet* 2016, **17**:693-703.
3. Morris SA, Daley GQ: **A blueprint for engineering cell fate: current technologies to reprogram cell identity**. *Cell Res* 2013, **23**:33-48.
4. Faure L, Soldatov R, Kharchenko PV, Adameyko I: **scFates: a scalable python package for advanced pseudotime and bifurcation analysis from single-cell data**. *Bioinformatics* 2023, **39**:btac746.
5. Huang S, Guo Y-P, May G, Enver T: **Bifurcation dynamics in lineage-commitment in bipotent progenitor cells**. *Dev Biol* 2007, **305**:695-713.
6. Mojtahedi M, Skupin A, Zhou J, Castaño IG, Leong-Quong RYY, Chang H, Trachana K, Giuliani A, Huang S: **Cell fate decision as high-dimensional critical state transition**. *PLoS Biol* 2016, **14**:e2000640.
7. Heinäniemi M, Nykter M, Kramer R, Wienecke-Baldacchino A, Sinkkonen L, Zhou JX, Kreisberg R, Kauffman SA, Huang S, Shmulevich I: **Gene-pair expression signatures reveal lineage control**. *Nat Methods* 2013, **10**:577-583.
8. Soldatov R, Kaucka M, Kastriti ME, Petersen J, Chontorotzea T, Englmaier L, Akkuratova N, Yang Y, Häring M, Dyachuk V, et al.: **Spatiotemporal structure of cell fate decisions in murine neural crest**. *Science* 2019, **364**:eaas9536.
9. Monod J, Jacob F: **Teleonomic mechanisms in cellular metabolism, growth, and differentiation**. *Cold Spring Harb Symp Quant Biol* 1961, **26**:389-401.
10. Richard A, Boullu L, Herbach U, Bonnafox A, Morin V, Vallin E, Guillemin A, Papili Gao N, Gunawan R, Cosette J, et al.: **Single-cell-based analysis highlights a surge in cell-to-cell molecular variability preceding irreversible commitment in a differentiation process**. *PLoS Biol* 2016, **14**:e1002585.
11. Cisló DJ, Delás MJ, Briscoe J, Siggia ED: **Reconstructing Waddington's landscape from data**. *Proc Natl Acad Sci USA* 2025, **122**:e2521762122.
- This work presents a formal computational geometry framework that reconstructs Waddington's landscapes by fitting Morse-Smale dynamical systems directly to high-dimensional single-cell data. The method models cell fate transitions as flows in a signal-dependent topography, accurately capturing developmental features like fixed points and basins of attraction.
12. Kamenev D, Kameneva P, Adameyko I: **The role of microheterogeneity in cell fate decisions in neural progenitors and neural crest**. *Curr Opin Neurobiol* 2025, **92**:103031.
13. Weinreb C, Rodriguez-Fraticelli A, Camargo FD, Klein AM: **Lineage tracing on transcriptional landscapes links state to fate during differentiation**. *Science* 2020, **367**:eaaw3381.
14. Herman JS, Sagar, Grün D: **FateID infers cell fate bias in multipotent progenitors from single-cell RNA-seq data**. *Nat Methods* 2018, **15**:379-386.
15. Huguet G., Magruder D.S., Tong A., Fasina O., Kuchroo M., Wolf G., Krishnaswamy S.: **Manifold Interpolating Optimal-Transport Flows for Trajectory Inference**; 2022: doi:(10.48550/ARXIV.2206.14928).
16. Schiebinger G, Shu J, Tabaka M, Cleary B, Subramanian V, Solomon A, Gould J, Liu S, Lin S, Berube P, et al.: **Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming**. *Cell* 2019, **176**:928-943.e22.

17. Weinreb C, Wolock S, Tusi BK, Socolovsky M, Klein AM: **Fundamental limits on dynamic inference from single-cell snapshots.** *Proc Natl Acad Sci* 2018, **115**:E2467-E2476.
18. De Belly H, Paluch EK, Chalut KJ: **Interplay between mechanics and signalling in regulating cell fate.** *Nat Rev Mol Cell Biol* 2022, **23**:465-480.
19. Yu VWC, Yusuf RZ, Oki T, Wu J, Saez B, Wang X, Cook C, Baryawno N, Ziller MJ, Lee E, et al.: **Epigenetic memory underlies cell-autonomous heterogeneous behavior of hematopoietic stem cells.** *Cell* 2017, **168**:944-945.
20. La Manno G, Soldatov R, Zeisel A, Braun E, Hochgerner H, Petukhov V, Lidschreiber K, Kastri ME, Lönnerberg P, Furlan A, et al.: **RNA velocity of single cells.** *Nature* 2018, **560**:494-498.
21. Lange M, Bergen V, Klein M, Setty M, Reuter B, Bakhti M, Lickert H, Ansari M, Schniering J, Schiller HB, et al.: **CellRank for directed single-cell fate mapping.** *Nat Methods* 2022, **19**:159-170.
22. Bergen V, Soldatov RA, Kharchenko PV, Theis FJ: **RNA velocity-current challenges and future perspectives.** *Mol Syst Biol* 2021, **17**:e10282.
23. Gorin G, Fang M, Chari T, Pachter L: **RNA velocity unraveled.** *PLoS Comput Biol* 2022, **18**:e1010492.
24. Wagner DE, Klein AM: **Lineage tracing meets single-cell omics: opportunities and challenges.** *Nat Rev Genet* 2020, **21**:410-427.
25. Bowling S, Sritharan D, Osorio FG, Nguyen M, Cheung P, Rodriguez-Fraticelli A, Patel S, Yuan W-C, Fujiwara Y, Li BE, et al.: **An engineered CRISPR-Cas9 mouse line for simultaneous readout of lineage histories and gene expression profiles in single cells.** *Cell* 2020, **181**:1410-1422.e27.
26. Li L, Bowling S, McGeary SE, Yu Q, Lemke B, Alcedo K, Jia Y, Liu X, Ferreira M, Klein AM, et al.: **A mouse model with high clonal barcode diversity for joint lineage, transcriptomic, and epigenomic profiling in single cells.** *Cell* 2023, **186**:5183-5199.e22.
27. Raj B, Wagner DE, McKenna A, Pandey S, Klein AM, Shendure J, Gagnon JA, Schier AF: **Simultaneous single-cell profiling of lineages and cell types in the vertebrate brain.** *Nat Biotechnol* 2018, **36**:442-450.
28. Spanjaard B, Hu B, Mitic N, Olivares-Chauvet P, Janjua S, Ninov N, Junker JP: **Simultaneous lineage tracing and cell-type identification using CRISPR-Cas9-induced genetic scars.** *Nat Biotechnol* 2018, **36**:469-473.
29. Alemany A, Florescu M, Baron CS, Peterson-Maduro J, van Oudenaarden A: **Whole-organism clone tracing using single-cell sequencing.** *Nature* 2018, **556**:108-112.
30. Kong W, Biddy BA, Kamimoto K, Amrute JM, Butka EG, Morris SA: **CellTagging: combinatorial indexing to simultaneously map lineage and identity at single-cell resolution.** *Nat Protoc* 2020, **15**:750-772.
31. Ratz M, von Berlin L, Larsson L, Martin M, Westholm JO, La Manno G, Lundeberg J, Frisén J: **Clonal relations in the mouse brain revealed by single-cell and spatial transcriptomics.** *Nat Neurosci* 2022, **25**:285-294.
32. Wang S-W, Herriges MJ, Hurley K, Kotton DN, Klein AM: **CoSpar identifies early cell fate biases from single-cell transcriptomic and lineage information.** *Nat Biotechnol* 2022, **40**:1066-1074.
33. Forrow A, Schiebinger G: **LineageOT is a unified framework for lineage tracing and trajectory inference.** *Nat Commun* 2021, **12**:4940.
34. Steinschaden T, Faure L, Soldatov R, Adameyko I: **A competition model of multilineage priming and cell-fate decisions.** *Cell Rep* 2026, **45**:116690.
35. Rodriguez-Fraticelli AE, Wolock SL, Weinreb CS, Panero R, Patel SH, Jankovic M, Sun J, Calogero RA, Klein AM, Camargo FD: **Clonal analysis of lineage fate in native haematopoiesis.** *Nature* 2018, **553**:212-216.
36. Erickson AG, Isaev S, Artemov A, He J, Semsch B, Murtazina A, Sun J, Mangold K, Chalou A, Frisen J, et al.: **Unbiased profiling of multipotency landscapes reveals spatial modulators of clonal fate biases.** *BioRxiv Prepr Serv Biol* 2025, <https://doi.org/10.1101/2024.11.15.623687>.
This publication introduces the clonal embedding approach called clone2vec and provides examples of how this approach helps to investigate clonal variation during embryonic development. clone2vec also facilitates the discovery of genes and regulatory mechanisms, which confound clonal variation. Importantly, this study also demonstrates the power of massive parallel perturbations of signaling pathways via Cas9/CRISPR together with integrated clonal information.
37. de Haan S, He J, Corbat AA, Belicova L, Ratz M, Vinsland E, Frisén J, Kelley MW, Andersson ER: **Ectoderm barcoding reveals neural and cochlear compartmentalization.** *Science* 2025, **388**:60-68.
Authors performed *in utero* nanoinjection of a barcoded lentiviral library at E7.5 for clonally resolved scRNA-Seq of the mouse ectoderm, neural crest, and otic placode. The study identifies distinct compartmentalization within the developing nervous system and cochlea, successfully linking differentiated cell types to their specific progenitors while resolving convergent differentiation pathways.
38. Bandler RC, Vitali I, Delgado RN, Ho MC, Dvoretzkova E, Ibarra Molinas JS, Frazel PW, Mohammadkhani M, Machold R, Maedler S, et al.: **Single-cell delineation of lineage and genetic identity in the mouse brain.** *Nature* 2022, **601**:404-409.
39. Keefe MG, Steyert MR, Nowakowski TJ: **Lineage-resolved atlas of the developing human cortex.** *Nature* 2025, **647**:194-202.
This study is a very rare example of lineage tracing using human primary tissue. The authors identify a pivotal midgestation transition where neurogenesis switches from glutamatergic to GABAergic. Furthermore, the study demonstrates that human truncated radial glia maintain a protracted neurogenic potential, generating late-born glutamatergic neurons that exhibit molecular features of deep cortical layers and contribute to the expansion of the subplate region.
40. Delgado RN, Allen DE, Keefe MG, Mancia Leon WR, Ziffra RS, Crouch EE, Alvarez-Buylla A, Nowakowski TJ: **Individual human cortical progenitors can produce excitatory and inhibitory neurons.** *Nature* 2022, **601**:397-403.
41. Zhu Q, Ge J, Liu Y, Xu J-W, Yan S, Zhou F: **Decoding anterior-posterior axis emergence among mouse, monkey, and human embryos.** *Dev Cell* 2023, **58**:63-79.e4.
42. Wagner DE, Weinreb C, Collins ZM, Briggs JA, Megason SG, Klein AM: **Single-cell mapping of gene expression landscapes and lineage in the zebrafish embryo.** *Science* 2018, **360**:981-987.
43. Weinreb C, Klein AM: **Lineage reconstruction from clonal correlations.** *Proc Natl Acad Sci USA* 2020, **117**:17041-17048.
44. Snippert HJ, van der Flier LG, Sato T, van Es JH, van den Born M, Kroon-Veenboer C, Barker N, Klein AM, van Rheenen J, Simons BD, et al.: **Intestinal crypt homeostasis results from neutral competition between symmetrically dividing Lgr5 stem cells.** *Cell* 2010, **143**:134-144.
45. Kinsler G, Fagan C, Li H, Kaster J, Dunne M, Vander Velde RJ, Boe RH, Shaffer S, Herlyn M, Raj A, et al.: **SpaceBar enables clone tracing in spatial transcriptomic data.** *BioRxiv Prepr Serv Biol* 2025, <https://doi.org/10.1101/2025.02.10.637514>.
Authors introduce SpaceBar, a novel cellular barcoding strategy that enables the simultaneous tracing and profiling of spatial transcriptomics at single-cell resolution. Applying this technology to melanoma xenografts, the authors reveal that spatial organization is the predominant driver of gene expression variation, though specific genes like SFRP1 exhibit high heritability within clonal populations.
46. Kamimoto K, Stringa B, Hoffmann CM, Jindal K, Solnica-Krezel L, Morris SA: **Dissecting cell identity via network inference and in silico gene perturbation.** *Nature* 2023, **614**:742-751.
47. Li C., Gao H., She Y., Bian H., Chen Q., Liu K., Wei L., Zhang X.: **Benchmarking AI Models for In Silico Gene Perturbation of Cells;** 2024: doi:[10.1101/2024.12.20.629581](https://doi.org/10.1101/2024.12.20.629581).